

Level 1 — Continuous: the CPSWF problem

space $L^2([0, R], r dr)$; $\mathcal{T} = B_R B_K B_R$ compact, self-adjoint, positive

$$\mathcal{T} \psi^{(m)} = \lambda_m \psi^{(m)}; \text{ CPSWFs } \psi^{(m)}; \lambda_0 \geq \lambda_1 \geq \dots > 0$$

EXACT — unitary equivalence, no approximation

$\Omega f = (\langle f, \varphi_{n,k} \rangle)_{k \geq 1}$: isometric isomorphism $L^2 \rightarrow \ell^2$

$\Omega \mathcal{T} \Omega^{-1} = B_K^{(B, \infty)}$; identical spectrum

Level 2 — Infinite discrete: the operator in Fourier-Bessel coordinates

space $\ell^2(\mathbb{N})$; $B_K^{(B, \infty)}$ bounded, self-adjoint, PSD; same eigenvalues λ_m (exact)

$$B_K^{(B, \infty)} v^{(m)} = \lambda_m v^{(m)}; v^{(m)} = \Omega \psi^{(m)} = \text{FB coefficient sequence of the CPSWF}$$

GALERKIN — truncation: the sole approximation step

$(B_K^{(B)})_{mk} = (B_K^{(B, \infty)})_{mk}$, $m, k \leq P$ (upper-left block)

$\lambda_m^{(P)} \leq \lambda_m$ (interlacing); $\lambda_m^{(P)} \nearrow \lambda_m$ as P grows

Level 3 — Finite: the finite radial prolate matrix

space $\mathbb{R}^P$; $B_K^{(B)}$ real symmetric, PSD, eigenvalues in $[0, 1]$

$$B_K^{(B)} c^{(m)} = \lambda_m^{(P)} c^{(m)}; \text{ finite radial prolate sequences } c^{(m)}$$

synthesis: $\psi_P^{(m)} = \sum_k c_k^{(m)} \varphi_{n,k}$ (finite radial prolate functions)

$\psi_P^{(m)}$ approximates $\psi^{(m)}$
(Theorems 5-7)